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Palaeomagnetism of the Southern Zone of the Lewisian (Precambrian) Foreland, NW Scotland

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Summary. Palaeomagnetic results are reported from 111 localities in metamorphic rocks from the Southern Zone of the Lewisian Complex and 12 sites from similar terrain on the island of Lewis and Harris. These rocks were magnetized during slow uplift following the *ca.* 1800 Ma Laxfordian tectonic/magmatic episode. The Southern Zone experienced diachronous uplift and there is a transition from predominantly positive NW remanence directions in the north to shallow negative NW directions in the south. More prolonged metamorphism in the south correlates with a transition from magnetite/sulphide to magnetite/hematite assemblages. The relict Ruadh Mheallan zone relatively unaffected by Laxfordian tectonism preserves a (A1) remanence $D = 82^\circ$, $I = 65^\circ$ which is sporadically recovered as high blocking temperature component in areas bordering this zone. Elsewhere, the migration of field directions is summarized as mean directions (A3) $D = 323^\circ$, $I = 44^\circ$, (A4) $D = 314^\circ$, $I = 14^\circ$ and (A5) $D = 313^\circ$, $I = -11^\circ$ which appear to follow on sequentially from the predominant A2 direction ($D = 286^\circ$, $I = 55^\circ$) observed in the Central Zone of the Lewisian Complex. The remanence directions are linked to a first approximation to the K-Ar hornblende ages and imply a migration of the ambient field direction through *ca.* 110° during an interval between 1×10^7 and $2 \times 10^8 \times$ yr at about 1600 Ma; the weight of the evidence suggests that the interval represented is between 0.5 and $1 \times 10^8 \times$ yr.

The Lewisian A2–A5 directions yield palaeopoles which follow on from 1800–1700 Ma magnetizations from the Hudsonian terrains of Greenland and North America and overlap with the youngest record from these regions on the pre-drift reconstruction; collectively the data define part of a large apw loop.

1 Introduction

The extension of palaeomagnetic studies to metamorphic terrains is defining a new body of data: in place of single pole positions representing discrete igneous or sedimentary events,

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these terrains yield a long-term record of the magnetic field behaviour over intervals of uplift and cooling (e.g. Morgan 1976a; McWilliams & Dunlop 1978; Piper 1980). Although the data cannot readily be dated with precision, they are providing a new conceptual approach to ancient apparent polar wander (apw) motions by providing an accurate record of the magnitude and direction of these motions in Precambrian times and necessitating a major revision of the apw paths. This study of part of the Lewisian metamorphic complex of NW Scotland is another example of the sequential acquisition of magnetizations over an interval of cooling; it extends the palaeomagnetic record from the Proterozoic 'Hudsonian' (*ca.* 1800 Ma) mobile terrains of the Laurentian Shield and contributes towards definition of the scale of apw motions in Middle Precambrian times.

2 Geology and age

The Lewisian complex forms the NW foreland of the Caledonian Belt (*ca.* 550–400 Ma) in Scotland and comprises a Central Zone (Fig. 1) preserving largely *ca.* 2900–2700 Ma (Badcallian) structures, and Northern and Southern Zones where these structures were largely reworked in Laxfordian times during an episode contemporaneous with the *ca.* 1800 Ma 'Hudsonian' (Churchill, Nagssugtoqidian, Ketilidian) mobile episodes of the Canadian Shield and Greenland. These two major divisions are conveniently separated by intrusion of the NW–SE Scourie dykes at 2400 Ma (Chapman 1979) into hot country rocks at depths of *ca.* 15–20 km (e.g. Tarney 1963; Dearnley 1973; O'Hara 1977). Tectonic aspects of the Laxfordian episode had terminated by about 1800 Ma (Watson 1975). The polyphase deformation and metamorphism, predominantly to amphibolite facies, during Laxfordian times was inhomogeneous even within these zones, and a large-scale pattern is defined by NW trending zones where deformation is alternately relatively weak and intense (Park 1973). Laxfordian deformation reached a climax about 1850 Ma (Watson 1975) but occurred within a period of protracted, and probably episodic, uplift and cooling from crustal depths equivalent to formation pressures of 15 kbar (30–40 km) in Badcallian times (O'Hara 1977) to cover thicknesses of 10–15 km by Laxfordian times, and with temperatures declining significantly only towards the end of the Laxfordian tectonism (Dickinson & Watson 1976; Watson 1977). This cover had been completely removed by the time of deposition of the Stoer Group (Torridonian) sediments at *ca.* 1000 Ma (Moorbath 1969; Watson 1975).

The two previous studies of the Lewisian Complex were confined to the Central Zone. Beckmann (1976) isolated a highly stable remanence by af demagnetization from nine dykes near Lochinver and a Scourian granulite; only samples from the Central Zone in the Lochinver/Assynt area possessed a high coercivity remanence and no meaningful remanence directions were recognized in bordering parts of the Northern Zone. Morgan (1976b) interpreted the Central Zone magnetizations in the context of slow cooling following the Laxfordian episode at *ca.* 1800 Ma. A similar interpretation was followed in a second palaeomagnetic study of the Central Zone by Piper (1979) which recognized several magnetizations (denoted A, B and C) of which the predominant A remanence ($D = 286^\circ$, $I = 55^\circ$) is identical to that found by Beckmann and broadly linked to K-Ar mineral ages averaging 1650–1625 Ma. Subsidiary magnetizations, of which 'C' is predominantly a group of loosely-defined low blocking temperature components, are believed to have been acquired during stages of the post-Laxfordian uplift from crustal depths of 6–10 km (Dickinson & Watson 1976).

The present study extends the coverage of the Lewisian metamorphic complex to the Southern Zone of the mainland and to the Island of Lewis. The collective results define a

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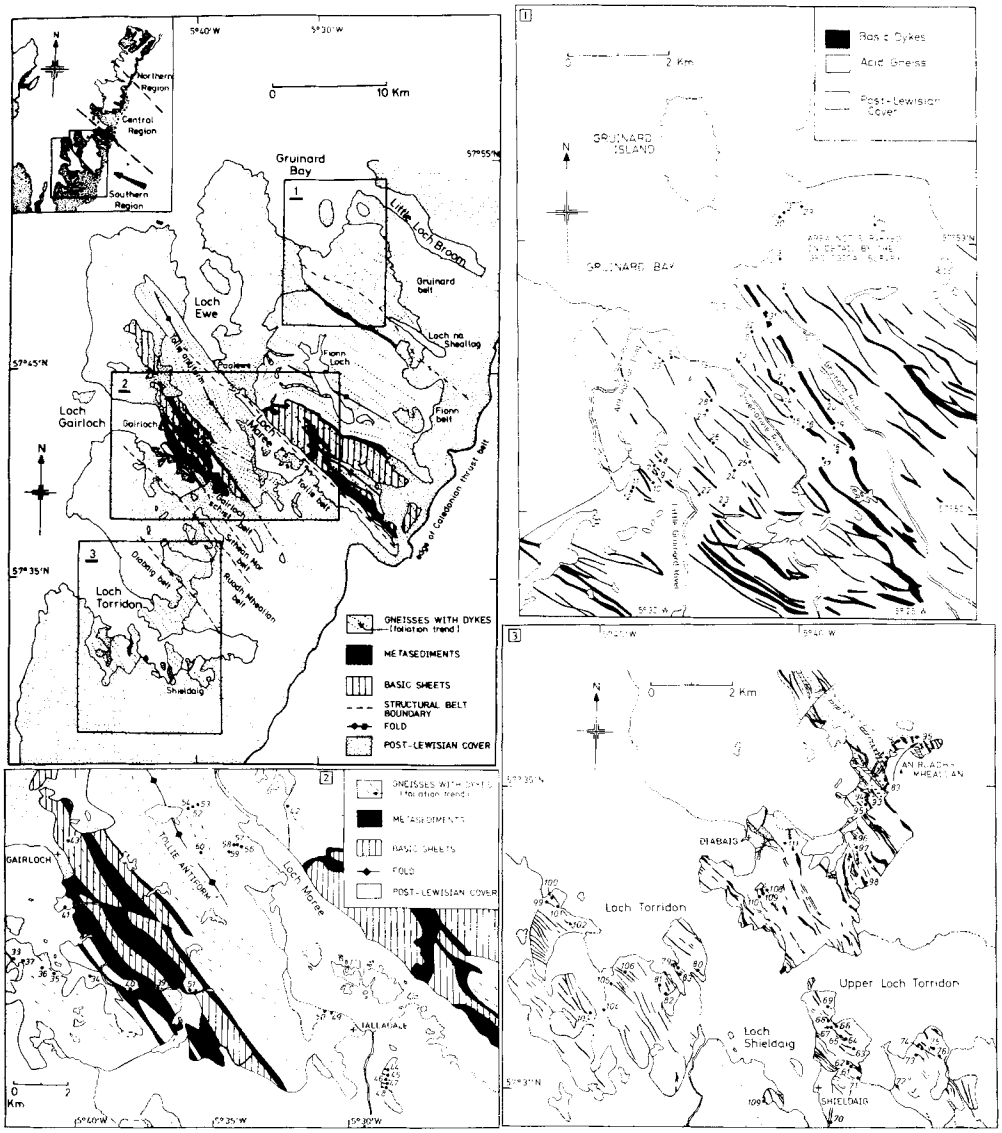


Figure 1. Geology of the Southern Zone of the Lewisian metamorphic complex, NW Scotland. Parts 1 to 3 show the distribution of the main lithological divisions within this zone in the Gruinard Bay, Gairloch and Torridon areas respectively, together with the palaeomagnetic sampling sites (numbered) of this study.

palaeomagnetic record which is largely distinct from that of the Central Zone of the Complex. In addition, new studies of the Upper Proterozoic Torridonian sediments (Smith, Stearn & Piper 1982) define with precision the configuration of the Hebridean Craton within the Laurentian Shield; as a result we are now able to consider the Lewisian results in the context of the Laurentian palaeopoles more precisely than has been possible hitherto and we show the Lewisian data define a segment of the Middle Proterozoic apw path for the Laurentian Shield.

The geology of the Southern Zone of the Lewisian Complex is described in the NW Highland Memoir (Peach *et al.* 1907). The northern sector of this zone, around Gruinard Bay, is predominantly a coarse acid gneiss containing numerous metamorphosed basic minor

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intrusions (Bowes, Wright & Park 1964). Further south in the Gairloch area these gneisses (in part granulites, the Ialltaig gneisses) are overlain by an extensive group of younger metasedimentary and meta-igneous rocks (the Gairloch schists and gneisses and the Loch Maree Group). These latter rocks display both Inverian and Laxfordian structures (Park 1964) and Park (1973) recognizes five phases of deformation in the Tollie gneisses. The area around Loch Torridon formed part of a classic geological study by Sutton & Watson (1951) which provided the basis of a chronological sub-division of the mainland Lewisian into the Scourian and Laxfordian episodes. Of particular relevance to this study, they recognized a small area known as the Ruadh Mheallan zone largely unaffected by the late Laxfordian metamorphic events and where NE trending structures contrast with the NW structures present elsewhere.

Moorbath & Park (1972) dated events in the Southern Zone using K-Ar mineral ages and identified three Laxfordian metamorphic events at *ca.* 1700, 1500 and 1400 Ma with a post-Laxfordian reheating event in the Torridon area at 1150 Ma; the collective data suggested a complex, and possibly diachronous, history of uplift. The Ruadh Mheallan Zone is largely undisturbed by this second metamorphism and yields a minimum age of *ca.* 1690 Ma compared with *ca.* 1480 Ma outside of this zone. Elsewhere during the third Laxfordian event at ca. 1400 Ma defined by K-Ar hornblende ages widespread retrogression occurred in the acid gneisses. (In this paper the term 'Laxfordian' is used to embrace the tectonothermal events during uplift of the Lewisian terrain prior to *ca.* 1150 Ma and post-dating the major *ca.* 1800 Ma tectonism.)

The Island of Lewis retains, in part, Badcallian (2900–2700 Ma) amphibolite facies gneisses (Watson 1975). It includes the South Harris tonalite-anorthosite-gabbro complex; granites and migmatites were, however, widely developed at late stages of the Laxfordian tectonism. Stages in the Laxfordian uplift are recorded by Rb-Sr mineral ages of 1654–1483 Ma (Lambert, Myers & Watson 1970) and K-Ar whole rock ages of 1720–1500 Ma (Lambert, Evans & Dearnley 1970).

3 Palaeomagnetic results

3.1 THE FIELD SAMPLE AND LABORATORY TREATMENT

The palaeomagnetic cores were drilled in the field and oriented prior to extraction by sights on the Sun and/or magnetic bearings in conjunction with topographic sights. Each palaeomagnetic site typically comprises six or seven independently oriented cores distributed over several metres of outcrop. The sampling was restricted to the basic units because these in general possess the least tectonic fabric and the highest ferrimagnetic content. Rock units with marked fabric were avoided both because the magnetic vector requires significant corrections for magnetic anisotropy (Beckmann, Olson & Sørensen 1977), and because the bulk anisotropy as determined in the laboratory may not reliably reflect the anisotropy of the components holding the stable remanence.

Thermomagnetic determinations (saturation magnetization J_s versus temperature) are listed in the data Tables (1–4) and suggest that magnetite is the predominant remanence carrier at most sites; detailed af demagnetization has accordingly been applied to the bulk of the collections. Each sample was progressively demagnetized in steps of 5–10 milliTesla (mT) up to a maximum field of 160 mT, or to fields at which the magnetizations could no longer be accurately measured or directional behaviour became erratic. The response of the magnetic vector to this treatment was frequently quite different for individual samples from the same site, and the selection of 'blanket' cleaning fields is not, in general, acceptable; our results are accordingly based on progressive af cleaning of each sample. Progressive thermal demagneti-

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zation was also undertaken on at least one sample from most sites in steps of 100° or 150°C to 450° or 500°C with incremental steps reduced to 20° or 30°C near the Curie point(s); some samples disintegrated above about 550°C following expansion of phyllosilicate phases.

Site mean directions of magnetization were calculated using a method similar to that of Dagley & Ade-Hall (1970): a consecutive pair of directions is taken from the measurement-demagnetization sequence for each core and the mean direction for the pair is calculated. All possible combinations of mean directions are then examined to determine the combination yielding the highest value of the Fisher precision parameter (k). In slowly-cooled metamorphic rocks, more than one component is frequently recognized in individual samples and the range of treatment covered by this calculation is first specified by inspection of the demagnetization plots.

In the Southern Zone of the Lewisian Complex samples were collected from 111 sites between Loch Broom and Loch Torridon (57.9° N to 57.5° N), and for purposes of discussion of results the Lewisian outcrop falls naturally into three areas (Fig. 1); these are: Gruinard Bay, Gairloch (including Tollie and the area south of Loch Maree) and Torridon (Fig. 1). On the Island of Lewis the Lewisian outcrop is predominantly granitic but the palaeomagnetic samples come from 12 localities in basic units including five sites across the South Harris tonalite-anorthosite gabbro complex (Dearnley 1963, and Fig. 6). The palaeomagnetic results from these sites are now outlined area by area.

3.2 GRUINARD BAY

Many of the 33 sites from the Gruinard Bay area come from basic to ultrabasic intrusions listed and described by Bowes *et al.* (1964). National Grid references for significant sites are listed in Table 1; thermomagnetic determinations identify magnetite as the magnetic phase although pyrrhotite is also identified at site 22 (Table 1). Samples exhibit a range of blocking temperature spectra: sites 13, 14, 22, 27 and 31 (Fig. 2) for example, have square shoulder thermal decay curves associated with high blocking temperatures in the range 550–600°C, while a smaller number of sites (e.g. 6, 18, 23) exhibit a low blocking temperature component at *ca.* 300°C. Af treatment in general produces a systematic movement in the vector from steep positive north/north-west to shallow north-west (e.g. 7, 14, 18, 23, 28 and 31, Fig. 2); a few sites (e.g. 13, 27 and 33) exhibit little movement of the vector but substantial fall in moment with af cleaning. Optimum grouping of site vectors is achieved by af treatment between 20 and 70 mT and defines directions not achieved by thermal treatment until 400–500°C; as observed by Morgan (1976c), the af demagnetization of magnetite-bearing metamorphic rocks can demagnetize a wider blocking temperature spectrum and selectively demagnetize more magnetic grains with lower blocking temperatures than with higher blocking temperatures. However, an example of the merits of thermal demagnetization is shown by sample from site 6 (Fig. 2) where it identifies an antiparallel component of magnetization between 580 and 600°C; we interpret this as a palaeofield reversal recorded by hematite or a hemo-ilmenite.

At all but three sites af cleaning improves within-site precision and, as a rule, significant movement of the magnetic vector is associated with low coercivity components removed by peak alternating fields up to 30 mT or temperatures up to 500°C; significant movement of the vectors does not in general occur over the defined critical blocking temperature ranges. We therefore believe that the af cleaned mean directions (Table 1) accurately represent magnetizations acquired during regional uplift and cooling following removal of a viscous component possibly associated with the Tertiary/Recent geomagnetic field. Of the 33 sites sampled in the Gruinard Bay area, nine were too weakly magnetized ($< 0.02 \times 10^{-5}\,Am^2kg^{-1}$)

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Table 1. Palaeomagnetic results for the Gruinard Bay area, Lewisian Complex, after af cleaning.

SITE	N_T	Grid reference	T_c	N	D	I	R	k	J_o ($\times 10^{-5}$)
2	6	NG 962930	-	3	*357.4	81.2	2.99	143	0.02 - 0.08
3	6	NG 961920	575	3	*254.6	52.5	2.64	5	0.26 - 0.63
4	6	NG 962914	560	3	59.5	74.5	2.99	283	1.53 - 56.78
6	6	NG 941901	570	6	*287.9	48.4	5.70	17	0.03 - 1.11
7	5	NG 937880	570	5	331.2	38.7	4.93	56	3.40 - 16.23
9	5	NG 937878	575	5	302.0	22.0	4.57	9	0.02 - 0.45
11	6	NG 932875	-	6	*335.1	26.6	5.02	5	0.005-0.021
12	6	NG 932875	-	6	321.5	-12.9	5.92	65	0.03 - 0.05
13	5	NG 953900	565	3	305.7	21.9	3.00	2444	0.06 - 0.35
14	5	NG 962892	570	5	308.1	5.4	4.99	292	3.18 - 6.55
15	7	NG 964889	-	4	*160.6	57.4	3.79	15	0.01 - 0.12
18	6	NG 972882	575	6	318.0	41.0	5.92	65	0.21 - 0.92
19	6	NG 972887	-	3	*317.1	-32.9	2.94	34	0.01 - 0.29
21	6	NG 962896	-	3	317.6	-5.5	2.98	132	0.01 - 0.08
22	6	NG 946874	330/570	6	326.1	12.0	5.79	24	0.30 - 5.71
23	5	NG 950872	580	4	300.2	6.0	3.86	21	3.28 -19.44
24	6	NG 952879	-	5	321.1	32.3	4.92	48	0.01 - 0.48
26	6	NG 948884	-	6	*300.9	30.9	5.94	66	0.02 - 0.06
27	5	NG 947890	575	5	315.4	52.3	4.85	27	0.79 -21.21
28	5	NG 948893	580	5	312.6	7.3	4.95	74	0.33 - 0.85
29	6	NG 966931	-	5	*197.7	-18.0	4.56	9	0.01 - 0.03
30	5	NG 961929	580	4	*349.3	53.6	3.96	78	0.52 - 1.05
31	5	NG 959909	575	5	315.9	27.2	4.99	711	0.63 - 3.40
33	5	NG 938900	585	5	340.0	40.3	4.99	366	5.11 - 6.88

N_T is the total number of samples considered within each site. Declination (D) and inclination (I) of the mean direction derived from N samples before and after af cleaning. R is the resultant vector, k the precision parameter and α_{95} the radius of the 95 per cent confidence circle about the mean direction. T_C is the Curie temperature (± 10°C) estimated from thermomagnetic curves and J_0 ($Am^2 kg^{-1}$) is the total NRM range of intensities. * denotes the af cleaned site mean directions excluded from the overall mean directions calculated for the southern zone of the Lewisian Complex.

to measure accurately, and means for the remaining 24 sites after cleaning are listed in Table 1.

3.3 GAIRLOCH AREA

Twenty-seven sites were sampled in amphibolitized NW–SE trending dykes in this region (Fig. 1) and National Grid references are listed in Table 2. In contrast with the Gruinard Bay area, these sites do not exhibit a wide variation in blocking temperature and are all associated with high blocking temperatures in the range 550–600°C. Hematite is only positively identified at site 44 but is evident from thermal demagnetization at a few other sites. An example of the variable response of samples from a single site to af cleaning is illustrated by site 46 (Fig. 3); an antiparallel component is present in the low blocking temperature fraction of

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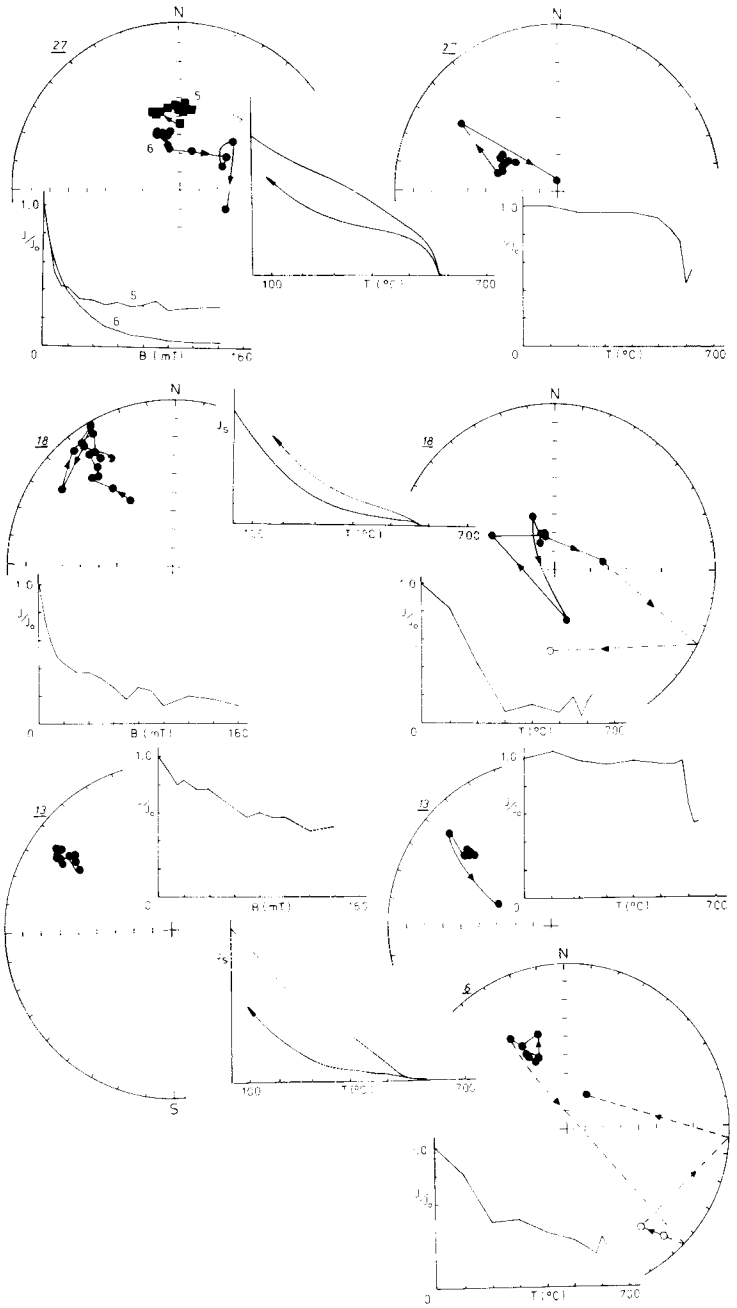


Figure 2. Behaviour of some representative directions and moments of magnetization of rocks from the northern (1 in Fig. 1) Gruinard Bay area with progressive af and thermal demagnetization, together with thermomagnetic curves. Projections are equal area, closed symbols are lower hemisphere plots and open symbols are upper hemisphere plots; J_0 is the initial (total NRM) intensity, and J_S the saturation remanence.

one sample; and each sample from this site achieved a stable end point at different stages of af treatment. At site 40 a southerly shallow (possible Palaeozoic) secondary component is removed by low fields (Fig. 3). Other sites such as 45 and 50 (Fig. 3) show little or no movement of the vector with either method of progressive treatment. An interesting example of a

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Table 2. Palaeomagnetic results for the Gairloch area, Lewisian Complex, after af cleaning.

SITE	N_T	Grid Reference	T_c		N	D	I	R	k	α_{95}	J_o ($\times 10^{-5}$)
34	9	NG 815725	580		9	99.4	79.8	8.94	131	4.5	0.13 - 0.62
36	6	NG 797728	-		5	-143.9	28.4	4.78	19	18.2	0.01 - 0.06
39	4	NG 840721	580		4	302.7	-7.6	3.99	461	4.3	0.07 - 0.13
40	5	NG 830720	580		5	326.7	-22.4	4.97	153	6.2	0.48 - 4.61
41	6	NG 804750	570	(i)	6	327.5	52.4	5.94	88	7.2	0.11 - 2.85
				(ii)	3	144.8	-53.5	2.94	33	21.9	
42	5	NG 887788	580		4	-211.6	3.7	4.00	1812	2.2	0.13 - 2.02
44	6	NG 924691	580/690		6	320.4	-8.2	5.97	149	5.5	2.24 - 3.76
45	6	NG 924690	580		6	315.5	-1.9	5.98	234	4.4	0.29 - 1.21
46	6	NG 924689	575		5	317.6	-12.6	4.99	521	3.4	0.62 -65.23
47	6	NG 924687	590		5	309.0	-17.3	4.97	144	6.4	1.14 - 9.81
48	6	NG 924685	575		6	335.0	58.8	5.95	108	6.5	0.77 - 1.97
49	5	NG 904712	570		4	316.0	-12.1	3.96	71	11.0	13.04-15.75
50	5	NG 901712	575		5	316.0	39.2	4.96	103	7.6	0.87 - 1.91
51	6	NG 853720	-		6	316.6	15.4	5.78	22	14.5	0.02 - 0.25
54	4	NG 851787	-		4	318.9	-3.4	4.00	793	3.3	0.04 - 0.23
55	5	NG 851787	-		4	328.9	17.2	3.99	362	4.8	0.04 - 0.23
56	6	NG 870773	-		3	315.8	-4.6	2.99	226	8.2	0.01 - 0.02
57	5	NG 869773	570		4	-347.4	54.0	3.00	234	6.0	0.08 -11.07
58	5	NG 867773	570		4	310.8	55.7	3.95	64	11.6	5.66 -14.37
59	5	NG 865770	-		3	-264.6	73.2	2.95	39	20.0	0.28 - 0.44
60	5	NG 855770	580		4	312.5	-17.2	3.99	553	3.9	0.50 - 6.02

Symbols as for Table 1.

reversal of magnetization within single samples is illustrated by site 41 (Fig. 3). The NW positive component is stable to af treatment to 30 mT and thermal treatment up to 550°C while an antiparallel component is recovered at higher fields and temperatures; paired cylinders from the same field core (e.g. 7 in Fig. 3) exhibit essentially the same behaviour with both af and thermal treatment; the transition takes place over approximately 50°C of thermal treatment with the higher blocking temperature component surviving to *ca.* 640°C. The decay of magnetic moments reflects the selective removal of two opposed magnetic vectors with a minimum intensity between 40 and 80 mT during the transitional movement between the two opposite polarities.

As in the Gruinard samples, the resultant effect of the cleaning treatment is to improve within-site precision and move some sites towards shallower NW inclinations although it has little or no effect on some other sites. Six of the 27 sites were too weak to measure accurately but site mean statistics for the remainder are listed in Table 2. They have a distribution which is non-Fisherian but forms a swathe of directions from steep positive NW to shallow negative NW (Section 4.1).

3.4 TORRIDON AREA

Samples were collected at 50 sites in basic dykes from this area at the grid references (for significant sites) listed in Table 3. The bulk of the thermomagnetic determinations (Table 3) again suggest that titanomagnetite close to pure magnetite in composition is the most impor-

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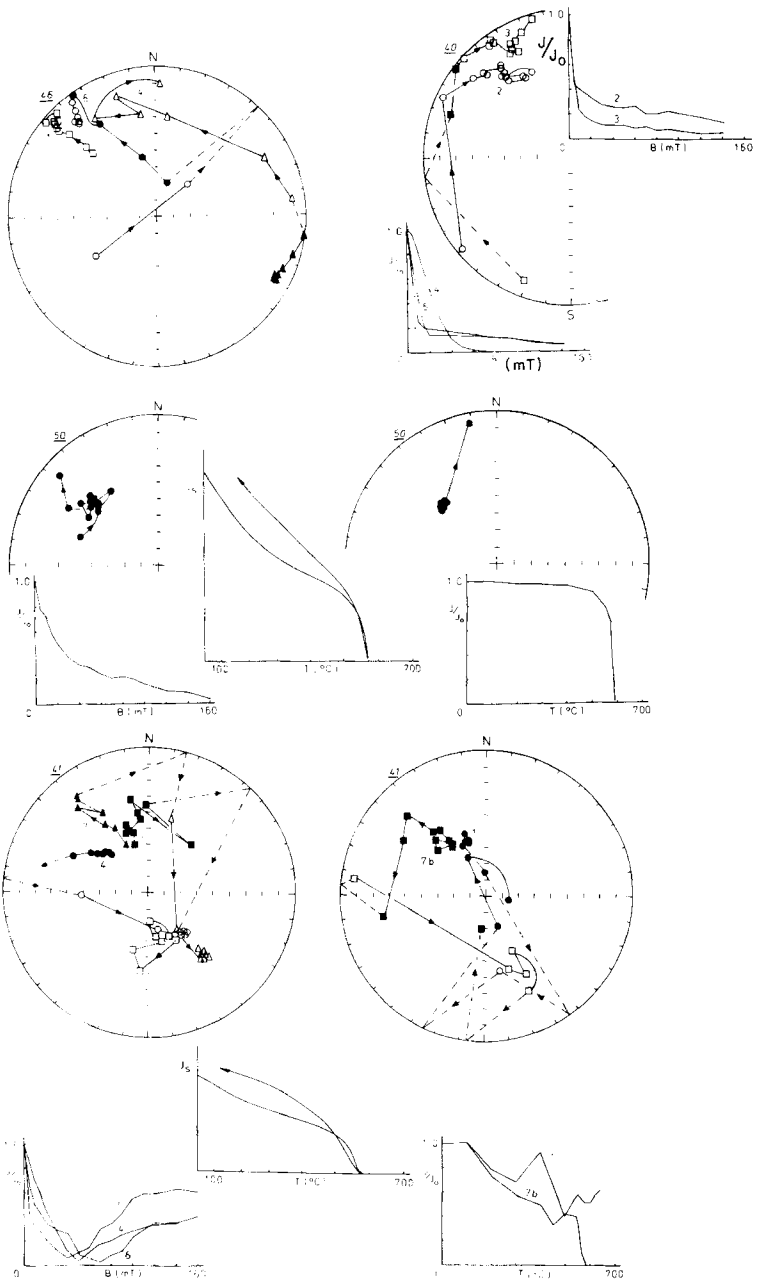


Figure 3. Behaviour of some representative directions and moments of magnetization of rocks from the central (2 in Fig. 1) Gairloch area with progressive af and thermal demagnetization, together with thermomagnetic curves. Symbols are as for Fig. 2.

tant remanence carrier; a few sites however (e.g. 62, 69 and 72) exhibit a second higher Curie point near 680°C indicating the presence of hematite. In addition site 107 shows a continuous increase in J_s on heating to about 190°C, an effect which Duff (1979) showed to be due to unsaturated hematite which is unblocked and aligned with the applied field.

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Table 3. Palaeomagnetic results for the Loch Torridon area, Lewisian Complex, after af cleaning.

SITE	N_T	Grid Reference	T_c	N	D	I	R	k	α_{95}	J_o ($\times 10^{-5}$)
61	7	NG 823543	-	3	•166.8	45.8	2.89	18	29.6	0.01 - 0.03
62	6	NG 822544	570/650	6	•282.0	-27.8	5.99	386	3.4	0.28 - 0.82
63	6	NG 824548	580	5	300.7	-4.6	4.91	45	11.5	0.38 - 0.66
64	6	NE 822550	-	6	301.7	-23.7	5.94	78	7.6	0.03 - 0.15
65	6	NE 821551	-	4	•283.4	-23.9	4.00	1328	2.5	0.03 - 0.08
66	6	NE 820553	590	5	64.0	61.7	4.97	121	7.0	0.05 - 0.22
67	5	NE 818553	580	3	•340.2	77.1	3.00	3940	2.0	0.09 - 0.42
69	6	NE 818558	580/640	5	•282.6	-9.0	4.95	79	8.7	0.35 - 3.88
70	6	NE 818529	585/680	6	301.7	-2.1	5.92	63	8.5	0.09 - 0.21
72	5	NE 835538	575/675	5	306.8	-19.9	4.93	55	10.4	0.46 - 0.82
73	5	NE 840546	580	(i)5	318.3	-17.1	4.94	72	9.1	1.08 - 3.01
				(ii)3	35.0	33.3	2.94	31	22.4	
74	5	NE 845545	580	5	•293.4	-0.5	4.96	95	7.9	0.21 - 0.62
75	6	NE 844547	590	3	305.9	-7.2	2.93	30	22.8	0.21 - 1.02
76	-	NE 841547			(conglomerate test, § 3.4)					
77	5	NE 802536	575	5	314.6	14.7	4.99	299	4.4	0.30 - 0.66
78	5	NE 781565	590	5	309.6	-10.5	4.91	45	11.6	0.37 - 3.57
79	5	NE 781567	590	5	318.6	-0.6	4.99	494	3.4	0.93 - 4.46
80	5	NE 786566	580	5	316.5	11.8	4.85	27	14.9	1.21 - 2.83
81	5	NE 776563	585	5	311.9	-12.8	4.89	37	12.7	2.46 -10.58
82	5	NE 778561	575	5	321.5	19.5	4.97	129	6.8	0.04 - 1.12
83	6	NE 833611	-	5	313.7	-7.7	4.96	99	7.7	0.01 - 0.03
84	6	NE 833612	-	4	326.2	0.7	3.89	28	17.8	0.01 - 0.08
85	5	NE 832615	570	4	114.2	60.3	4.00	792	3.3	0.06 - 0.58
87	5	NE 833618	570	4	312.9	18.9	3.99	209	6.4	0.19 - 2.36
88	4	NE 835621	580	4	78.7	51.7	3.98	187	6.7	12.19 -69.64
89	5	NE 836622	580	5	* 40.9	0.5	4.95	89	8.1	8.54 -92.98
91	5	NE 839622	590	3	84.9	55.1	2.93	30	22.8	1.67 -77.66
92	5	NE 840622	-	3	73.7	67.9	2.98	118	11.4	0.04 - 0.16
93	5	NE 828608	580	5	298.8	-3.9	4.96	105	7.5	0.03 - 0.10
94	5	NE 828607	570	5	312.6	42.1	4.97	116	7.1	0.40 - 1.04
97	6	NE 825596	580	6	•283.8	-26.3	5.94	80	7.5	0.04 - 0.14
98	5	NE 827587	575	3	•125.5	22.9	2.92	24	25.8	0.02 - 0.08
99	5	NE 748583	580	5	303.5	18.2	4.97	130	6.7	2.74 - 5.65
100	5	NE 750584	580	5	306.1	12.7	4.88	33	13.6	0.14 - 9.35
101	5	NE 751582	585	5	311.7	20.5	4.93	62	9.8	6.48 -13.67
102	5	NE 754577	575	5	309.3	-2.0	4.94	65	9.5	0.14 - 3.82
103	6	NE 760556	580	5	317.8	11.8	4.97	119	7.0	0.16 - 1.50
104	5	NE 763557	-	5	311.3	8.3	4.98	192	5.5	0.74 - 3.41
105	3	NE 764563	580	3	322.3	31.4	2.99	185	9.1	0.11 - 6.44
106	4	NE 764563	-	4	•340.7	21.9	3.99	481	4.2	0.11 - 6.44
107	5	NE 767565	590	5	321.6	13.8	4.96	109	7.4	2.27 -33.40
108	6	NE 802585	575	5	315.2	-10.6	4.93	263	4.7	0.09 - 2.14
109	5	NE 802584	-	5	309.4	-7.0	4.97	151	6.2	0.17 - 0.40
110	6	NE 801584	-	6	323.7	-30.0	5.81	27	13.2	0.01 - 0.02

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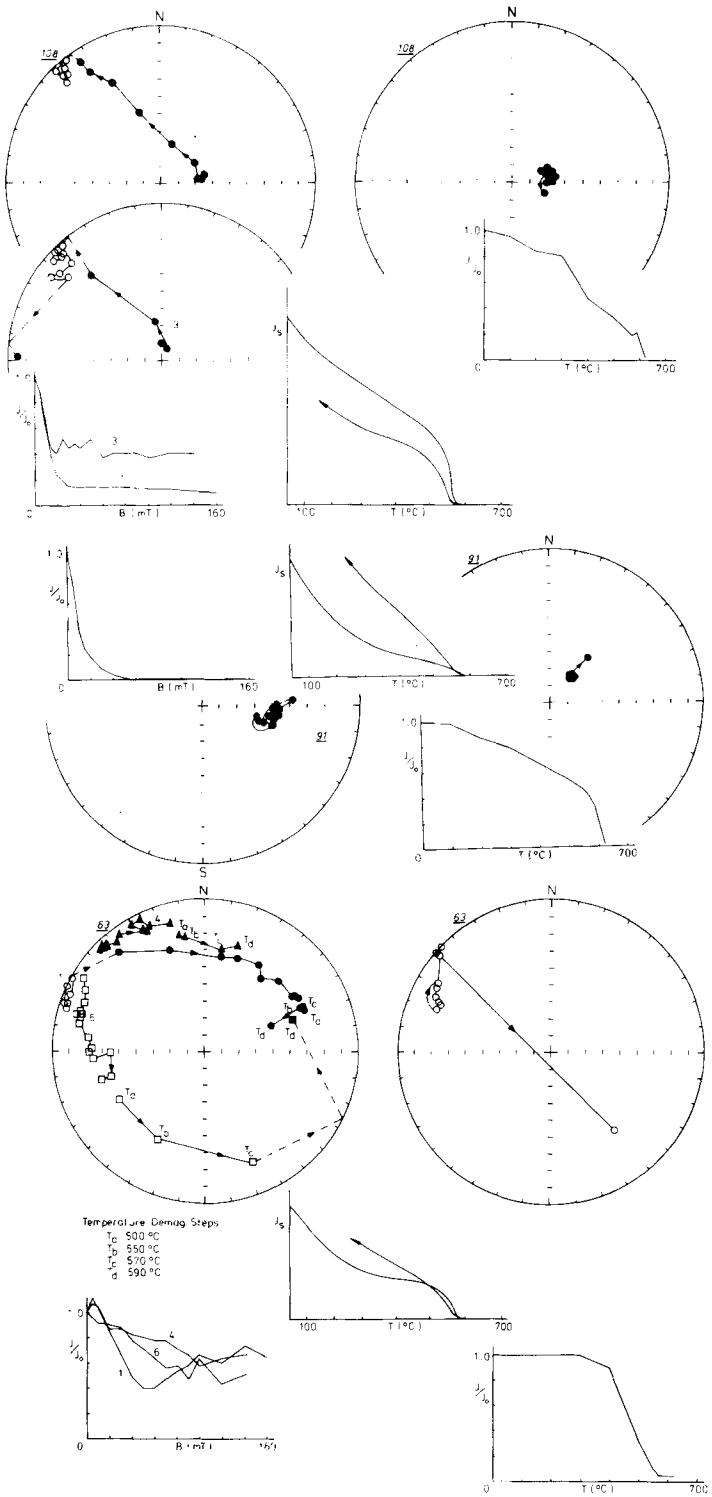


Figure 4. Behaviour of some representative directions and moments of magnetizations of rocks from the southern (3 in Fig. 1) Torridon area with progressive af and thermal demagnetization, together with thermomagnetic curves. Symbols are as for Fig. 2. Site 91 is from the relict Ruadh Mheallan zone.

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Most sites exhibit square shoulder J/J_0 curves with thermal demagnetization suggesting a narrow range of blocking temperatures at 570–590°C, and at most sites af cleaning produces little change in remanence direction. An exception is site 108 (Fig. 4) where af treatment identifies a large migration to a stable end point which is not achieved by thermal demagnetization and probably resides in a very narrow blocking temperature range. Five sites in the south side of Loch Torridon (63, 69, 73, 82, 105) exhibit systematic movement of the remanence vector with af treatment from shallow NW towards a positive westerly direction. In sample 63–1 (Fig. 4) this stable end point is reached with af treatment alone, but in some other samples with af and thermal treatment the moments dropped below the limits of accurate measurement before this end point could be achieved. In other samples (Fig. 4) thermal treatment alone is unable to reveal this high coercivity direction but it can be successfully recovered in conjunction with af treatment. Park (1975) suggests that whilst af demagnetization adequately removes less stable magnetizations it is unable to discriminate between magnetizations of higher stability because of the overlapping coercive force spectra; however, because the peaks of the blocking temperature spectra are distinct, subsequent thermal treatment can successfully discriminate between these magnetizations. At site 69 this high blocking temperature component might reside, in part, in hematite but of the other sites we found no evidence for hematite and believe that both components reside in magnetite. At other sites where hematite is present (Table 3) no specific magnetizations are associated with this mineral.

Ten palaeomagnetic sites (83–92) were sampled across the relict Ruadh Mheallan zone. Two sites proved too weak to measure but of the remaining eight sites, five yielded a univectorial easterly positive direction of magnetization distinct from the other five sites of this study (Figs 4 and 7). Both thermomagnetic determinations and thermal demagnetization confirm that this component resides in magnetite (Table 3). Since this remanence direction predominates in this relict massif of Scourian rocks little modified in Laxfordian times (Sutton & Watson 1951) we suggest that it is a somewhat older remanence than the NW intermediate positive to shallow directions which predominate elsewhere. We provisionally link it with the isolated NE directions recorded in the terrains affected by Laxfordian reworking notably site 4 from Gruinard Bay, site 34 from Gairloch and the high blocking temperature component noted above from sites on the south side of Loch Torridon. Whilst the entire Lewisian terrain would have been at high temperatures during the Laxfordian episode (Dickinson & Watson 1976) effects such as paucity of volatiles might have produced local variations in the regional cooling and hence in the age of magnetization.

3.4 CONGLOMERATE TEST, LOCH TORRIDON

Two cores were drilled in seven Lewisian boulders (amphibolitized basic facies) in a conglomerate at the base of the Torridonian Sandstone on the south side of Loch Torridon (site 76). The Lewisian basement was being actively degraded during deposition of the Torridonian Sandstone (Stewart 1975) at *ca.* 800–1000 Ma (Moorbath 1969), and the basal conglomerate provides suitable material for testing the long-term magnetic stability of the Lewisian metamorphic rocks. Total NRM intensities of these cores ranged widely from 4.6 to $0.02 \times 10^{-5}\,\mathrm{Am^2\,kg^{-1}}$. Following identical af treatment to the remainder of the collection optimum mean directions are consistent within individual boulders (Fig. 5) although the pebble magnetizations are statistically random (resultant vector, $R = 3.17$ for $N = 7$ boulders) at the 95 per cent confidence level, with directions unrelated either to the Lewisian basement or the Torridonian sediments (Stewart & Irving 1974). This test effectively shows that the Lewisian rocks have not been appreciably remagnetized since Upper Proterozoic times.

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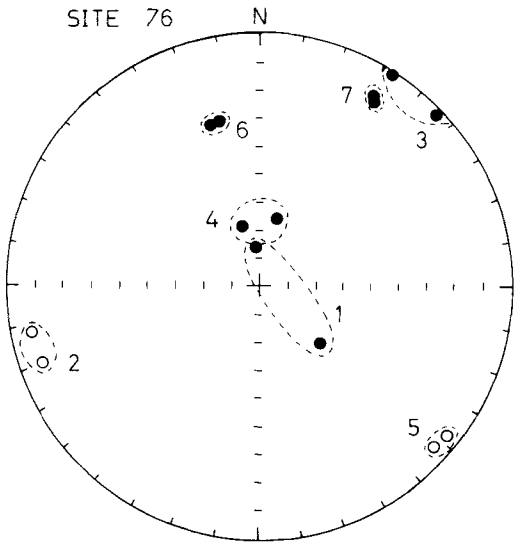


Figure 5. Directions of magnetization after af cleaning treatment of separate cores from seven boulders in a conglomerate at the base of the (*ca.* 1000 Ma), Torridon Group in the Torridon area.

3.5 ISLAND OF LEWIS AND HARRIS

Sampled localities and examples of demagnetization behaviour from the Island of Lewis are illustrated in Fig. 6. Site 1 from anorthositic facies near Butt of Lewis and sites 2 and 3 from ultrabasic dykes have predominantly weak components of magnetization defining shallow southerly directions of magnetization with treatment in low fields; one example exhibits a complete reversal above 90 mT. Although retained to about 550°C in some samples this component is only the lowest blocking temperature fraction in some samples: two examples migrate to a steeper negative direction (mean $D = 162°$, $I = -57°$) and most examples from

Table 4. Palaeomagnetic results for Lewisian rocks, Island of Lewis and Harris.

Site	T_c	N	D	I	R	k	J_c ($\times 10^{-5}$)
1	570	6	186	-27	5.78	22	0.02
2	580	6	202	-15	5.83	30	3.42–0.41
3	300,575	6	154	-11	5.98	22	0.41–0.08
4	565	5	171	-70	4.82	22	1.57–0.06
5	555	9	330	3	8.84	50	0.03
6	570	12	197	-45	11.84	73	0.05–0.02
7	570	9	125	-57	8.80	40	7.28–0.70
8	330,565	8	309	3	7.51	14	14.65–0.96
9 (i)	575	6	165	7	5.91	59	8.70–0.04
(ii)		6	317	-16	5.93	77	
11	660	8	63	73	7.96	167	0.15–0.02
12	580	6	293	16	5.60	12	0.82–0.40

Symbols as for Table 1; site locations are shown in Fig. 6.

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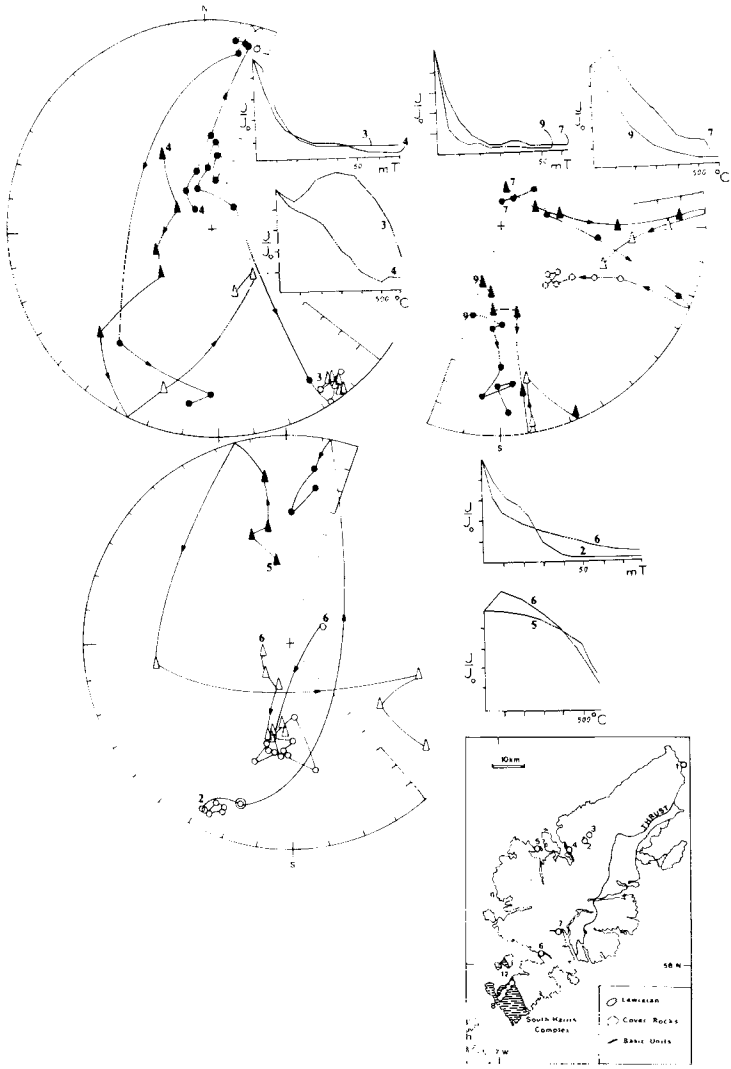


Figure 6. Behaviour of some Lewisian metamorphic rocks from the Island of Lewis and Harris, Outer Hebrides, with progressive af (circles) and thermal (triangles) demagnetization; symbols are as for Fig. 2. The inset map shows the location of the sampling sites (Table 4) in this area.

site 3 migrate to a steep poorly-defined direction ($D = 46°, I = 70°$) at highest stages of treatment. At site 4 another amphibolitized basic unit exhibits two low coercive force components; one (0–20 mT) is probably associated with the Tertiary/Recent field and is not listed in Table 4, a second (30–80 mT) is comparable to the initial component at sites 2 and 3 and two examples exhibit a complete reversal of polarity with progressive treatment; this direction survives thermal treatment to 530°C, above which directions migrate to a steeper negative direction (Table 4) comparable to the highest coercive force component from two samples at site 2. Site 5 from a basic body on the south side of the Loch Roag has two components: a shallow component exhibits complete reversal with progressive thermal treatment (Fig. 6) while a highest coercive force component possesses a steeper inclination. Sites 6 and 7 from amphibolitized basic bodies on either side of Glishan migrate to intermediate negative

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directions at the highest stages of treatment following progressive removal of a steep positive component.

Sites 8 to 12 are from anorthositic and metagabbro facies of the South Harris Complex. Site 8 exhibits Curie points due to both pyrrhotite and magnetite, and a remanence stable at lower af fields and temperatures (Table 4) probably resides in the former mineral; erratic directional behaviour takes place at higher stages of treatment and no stable component was clearly attributed to the magnetite. At site 9 the strongly magnetized samples ($>3.5 \times 10^{-5}$ Am^2kg^{-1}) define direction (1) in Table 4 while the remaining weak samples migrate to a less well-defined and broadly antiparallel direction comparable to site 8. Thermal demagnetization of site 11 produces a large fall in moment above 300°C and thermomagnetic determinations show a large initial fall in J_s with a single Curie point at 660°C attributable to hematite; it is probable that maghemite is the remanence carrier here because this mineral converts continually to hematite when heated in air and for this reason the remanence (Table 4) is unlikely to be of the same age as the other sites. At site 12 stable end points with westerly declinations have an elongated distribution of inclinations; magnetite is the carrier here.

While a NW shallow component of remanence (sites 5, 8, 9 and 12) comparable to the Southern Zone is present on Lewis and Harris, a southerly shallow to negative remanence with locally reversed polarities and a tendency to move to steeper negative directions at the highest stages of treatment (sites 1, 2, 3, 6 and 9) is also present here. It resides in magnetite and probably represents a stage in the *ca.* 1650–1480 Ma uplift history (Lambert *et al.* 1970). More detailed sampling of this area is necessary to evaluate its significance.

4 Discussion of results

4.1 THE MAGNETIC RECORD OF THE SOUTHERN ZONE

Within the individual (Gruinard, Gairloch and Torridon) areas there is no specific correlation between site mean direction and blocking temperature, structural level or geographic location (Smith 1980). However, when the Southern Zone is considered as a whole, a systematic variation is identified from area to area (Fig. 7): in Gruinard Bay site mean directions are distributed along a swathe from NW steep positive to shallow positive/negative; a similar distribution is recorded in the Gairloch area but here the majority of sites tend to be concentrated at shallow negative inclinations. Further south in the Torridon area NW steep positive directions are largely absent and most sites have shallow positive to negative inclinations. There is thus a trend in magnetization directions across the Southern Zone from NW intermediate positive directions in the north (or close to the direction which predominates in the Central Zone, Beckmann 1976; Piper 1979) to NW shallow negative directions in the south. Since no systematic variation in magnetic properties is identified here, the most simple explanation for this observation is that the region experienced diachronous uplift and cooling. This pattern is complicated, as noted above, by the sporadic preservations of older remanence directions notably in the relict Ruadh Mheallan belt.

4.2 AGE OF THE SOUTHERN ZONE MAGNETIZATIONS

It is difficult to assign a precise age to these magnetizations because under conditions of elevated temperatures followed by slow cooling, the critical range of blocking temperatures is considerably lowered (Pullaiah *et al.* 1975). Blocking temperatures of 450°C for hematite and 380°C for magnetite, with further reductions to 300 and 200°C respectively due to titanium substitution, have been suggested for cooling periods of 1 Ma (Pullaiah *et al.* 1975). Temperatures of this order are inferred for late stages of the Laxfordian episode when

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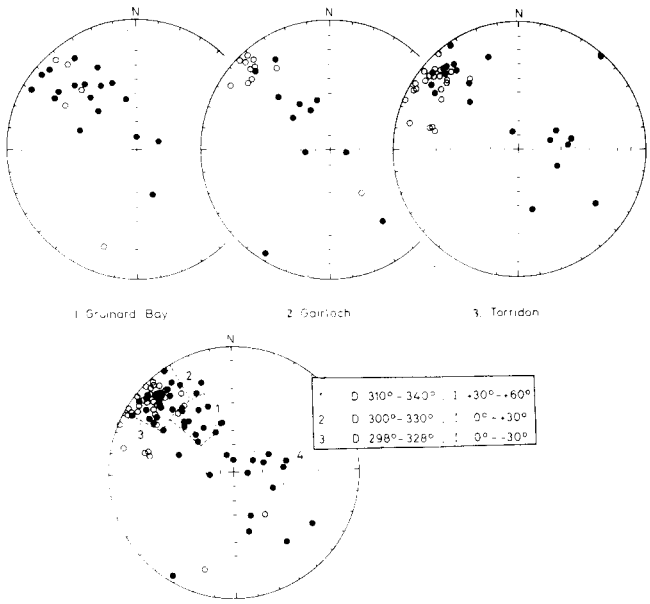


Figure 7. Site mean directions of magnetization after cleaning from the Gruinard Bay (northern) Gairloch (central) and Torridon (southern) areas of the Southern Zone of the Lewisian Complex. The combined plot of all mean directions shows the subdivisions that have been made to compute representative palaeomagnetic poles (A_1, A_3–A_5) from this area.

present levels were buried at *ca.* 10 km, but to determine the magnetization age precisely we would need to know the *rate* of uplift and cooling. In fact, there is some circumstantial evidence to suggest that periods of nearer 1 Ma than say 10 Ma were involved in cooling through the critical range of blocking temperatures, because single reversals of magnetization are sometimes recorded in individual samples but in no case is a complete double reversal of palaeofield observed. (As noted by Morgan 1976b, the fact that most samples possess a strong univectorial remanence implies that the palaeofield during uplift and cooling was predominantly of one polarity; the relatively small number of our sites showing a reversal would tend to confirm this.)

In the Torridon and Gairloch areas the K-Ar dates of Moorbath & Park (1972) fall into well-defined groups attributed to separate metamorphic events rather than to lateral changes in cooling, and this precludes a precise correlation with the magnetizations. Amphibolitized dykes yield K-Ar (hornblende) ages of 1504–1468 Ma in the Torridon (Diabaig) area and 1472 Ma in the Gairloch area. Older dates of 1712–1662 Ma have been obtained from less foliated dykes including those of the Ruadh Mheallan zone. The third Laxfordian episode at *ca.* 1400 Ma involves metamorphic retrogression in the acid gneisses but the basic rocks seem to be little affected, and younger ages of 1160 and 1148 Ma on chloritized biotites from acid gneisses in the Torridon area are attributed to localized reactivation along faults and shear belts (Moorbath & Park 1972); it is thus unlikely that these younger events are relevant to the high blocking temperature components. However, K-Ar hornblende ages of 1690–1678 Ma on unfoliated basic dykes and 1723–1668 Ma on regional gneisses in the Gruinard Bay area correspond to the first Laxfordian event recognized further south (muscovite dates here are 1516–1478 Ma), but they are significantly older than the Diabaig and Gairloch dykes and consistent with the view that the NW intermediate directions (which predominate here) are older than the shallower directions which predominate in the Torridon area; the

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latter is also where the later Laxfordian events were more intense. In the Gairloch area where the third Laxfordian episode (LD3) is well developed (Park 1964), the directions of magnetization are predominantly of negative inclination and this is interpreted as the youngest palaeofield direction. Our magnetic studies illustrate another aspect of this contrasting cooling history because hematite proves to be absent in the north and much more widely developed in the south (Tables 1–3 and Section 3).

Although four of the sites yielding easterly positive directions of magnetization come from the Ruadh Mheallan zone, three sites within this zone (83, 84 and 87) yield shallow NW directions typical of the Torridon and Gairloch areas. In the context of these results this area would appear to merit more detailed study because some of the larger dykes are amphibolitized at the margins but preserve a primary igneous texture in their interiors (Sutton & Watson 1951); also a (hornblende) K-Ar age of 2098 Ma from the margin of one of the largest dykes contrasts with a (whole rock) determination of 2230 Ma from the same dyke (Moorbath & Park 1972). However, of more relevance to the magnetization age are K-Ar determinations ranging from 1880–1681 Ma from the amphibolitized basic units. We correlate high blocking temperature magnetizations here with the second Laxfordian episode of amphibolite facies metamorphism (LM2 of Moorbath and Park) or immediately later stages of cooling, and believe that they predate the LM3 or retrogression in the acid gneisses with production of low temperature assemblages (chlorite/muscovite). As noted in the following sections, this is confirmed by a broad correlation of these poles with *ca.* 1800–1600 Ma poles. The LM3 metamorphism is linked to the LD3 and LD4 deformation episodes of which the former involves major rotations of dykes in upright folds (Park 1964, 1973). The maximum age of this episode is dated 1700–1670 Ma by reference to pegmatite ages (R. G. Park, private communication), which is therefore a likely maximum estimate of the age of these uplift-related magnetizations. The age assignments in Fig. 8 indicate the range of late LM1 or LM2 hornblende ages but they cannot be regarded as specific ages of these

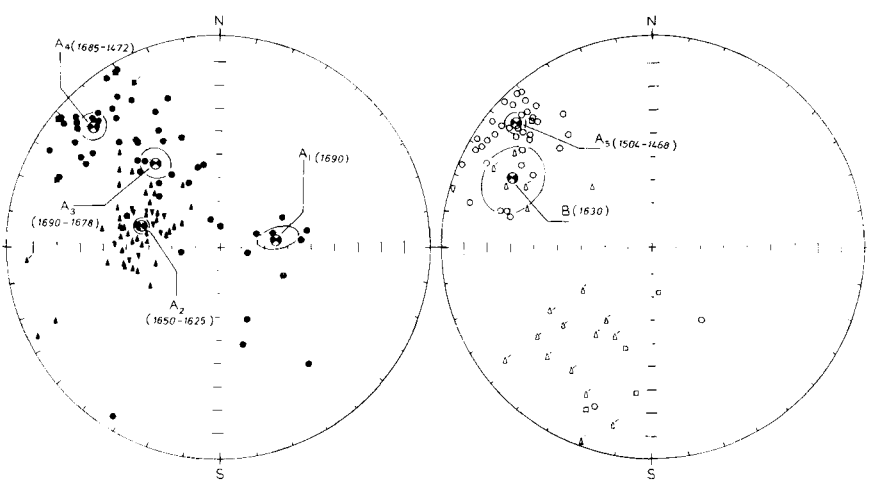


Figure 8. Summary of all site mean directions of magnetization from the Lewisian complex together with representative mean palaeomagnetic directions (equivalent to poles A_1 to A_5 and B, see text) and α_{95} confidence circles. The assigned ages are K-Ar or Rb-Sr uplift (amphibole) ages from the respective units yielding the palaeomagnetic directions and should only be taken as approximate indicators of the magnetization age. Circles are mean directions from the Southern Zone (this study), triangles are mean directions from the Central Zone (Piper 1979; Beckmann 1976) and squares from the Islands of Lewis and Harris (this study); symbols indicated by a dash are antiparallel directions.

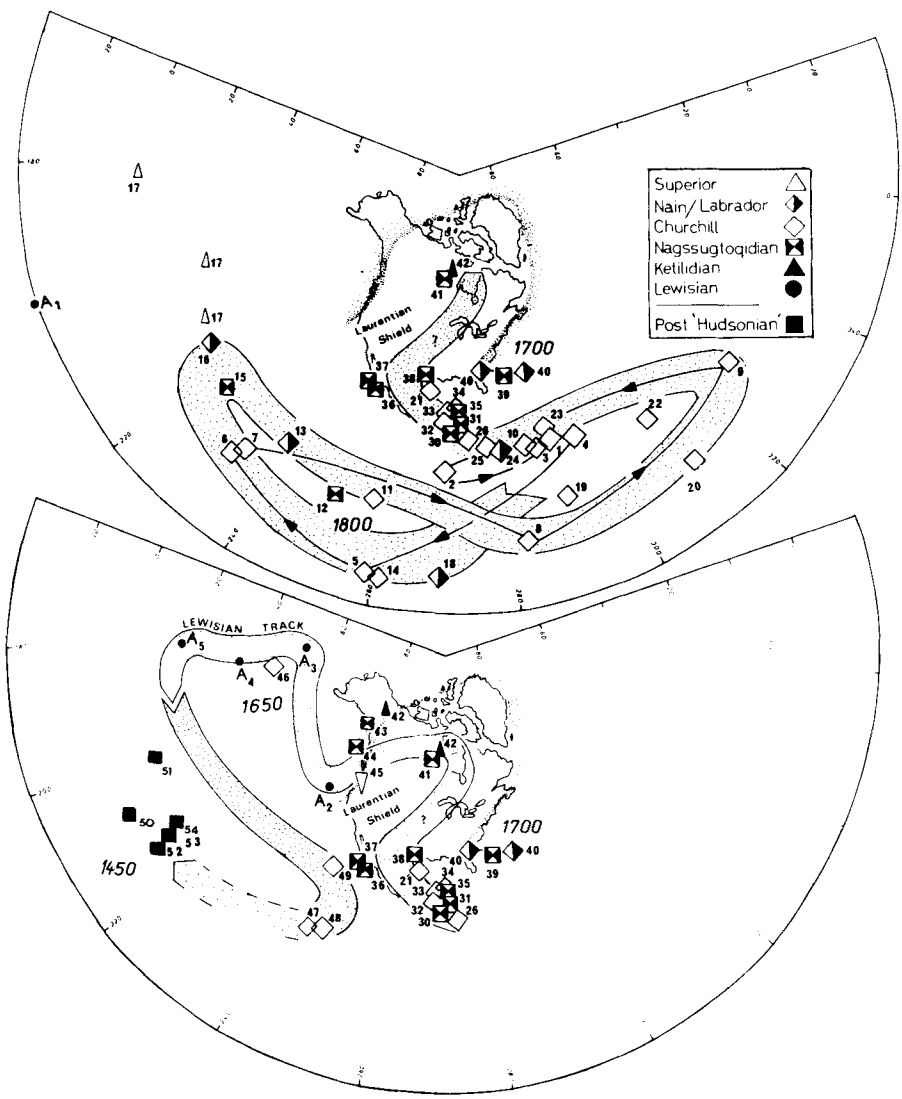
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magnetizations. In the context of the geological history, they are more likely to be representative of the steeper directions in the Gruinard and Gairloch areas largely or wholly unaffected by the later deformations. The coherent directions in the Torridon area would appear to be syn- or post-LD3 since they are dispersed by correction for this deformation.

4.3 PALAEOMAGNETIC MEAN POLES

We group the seven sites defining easterly positive directions in Table 3 to derive a mean $D = 82.1°$, $I = 65.4°$ ($N = 7$ sites, $R = 6.8$, $k = 40$, $\alpha_{95} = 9.7°$). This direction is (arguably) older than the other directions of this study and referred to as A1; it corresponds to a palaeomagnetic pole at 59.0° E, 42.3° N ($dp = 12.8°$, $dm = 15.7°$).

The remaining site directions define a non-Fisherian distribution mostly confined to a swathe of less than 30° width but ranging in inclination from +60° to −30° (Fig. 8). To calculate group means the site means were grouped into those falling within equal areas of



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the stereonet of size 30° (Fig. 7). The calculated means include 83 per cent of the sites with NW declinations and yield the following means:

	N	R	k	α_{95}	D	I	Palaeomagnetic pole	(dp dm)
Group 1 (A3)	11	10.8	50	6.5	322.7	+44.4	48.9° N 230.2° E	(5.1°, 8.2°)
Group 2 (A4)	20	19.7	61	4.2	314.0	+14.4	28.4° N 228.6° E	(2.2°, 4.3°)
Group 3 (A5)	27	26.6	60	3.6	312.5	−10.6	16.2° N 224.3° E	(1.9°, 3.7°)

D is the mean declination and I the mean inclination of the direction of magnetization. N, R, k and α_{95} are the Fisher statistics explained in the footnotes of the palaeomagnetic tables. dp and dm are the semi-axes of the 95 per cent confidence oval about the palaeomagnetic pole in the colatitude direction and at right angles to it, respectively.

Since these directions are believed to follow sequentially from the high blocking temperature remanence (A2) identified in the Central Zone (Beckmann 1976; Piper 1979, Section 4.1) they are referred to as A3 to A5, and represent a segment of the apw path some time between *ca.* 1690 and 1470 (?) Ma but we have no precise indication of the time period involved. Indeed, the distribution of all palaeomagnetic directions derived from af/thermal cleaning from sites in the Lewisian complex (Beckmann 1976, Piper 1979 and this study) as illustrated in Fig. 8, defines a swathe of directions running from the mean A1 direction to the mean A5 direction; we interpret this as defining a migration of the magnetic field relative to the region. The ages assigned to the mean poles A3 to A5 in Fig. 8 are the ranges of K-Ar hornblende ages from the three divisions of the Southern Zone and are only very provisional estimates for the ages of magnetization. We can only conclude from these data that the field migration occupied a period between 10^7 and 10^8yr, but, as noted in the next section, comparison with other Laurentian Shield data permits some refinement of this estimate.

Sporadic shallow E-W magnetizations occur in the Central Zone of the Lewisian Complex

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Figure 9. Summary of palaeomagnetic poles from the Laurentian Shield assigned to the approximate interval 1850–1600 Ma. The poles from Greenland and NW Scotland are corrected for Phanerozoic drift (Bullard *et al.* 1965 and text).

These data are predominantly from the terrains mobilized or overprinted by the *ca.* 1800 Ma Hudsonian mobile episodes and are accordingly linked to their structural provinces. The poles are summarized by Roy & Lapointe 1976, Cavanaugh & Seyfert (1977), Irving & McGlynn (1979) and Piper (1979) and (with their assigned ages in brackets) are from: 1, Kahocella primary (1873); 2, Seton Formation 'A' magnetization (1831); 3, Upper Gibraltar Formation; 4, McLeod Formation; 5, Douglas Peninsular Formation 'A' magnetization; 6, Stark Formation (1845–1630); 7, Totachwi Formation (1845–1818); 8, 9, 10, Pearson Formation, A, B and C magnetizations (1→ 10 is stratigraphic sequence in Athapuscow geosyncline); 11, Takiyuala Formation; 12, Nordfjord gneisses (1950–1750); 13, Indian Harbour dykes; 14, Peninsula sill (1800); 15, Nag. 2 hbt magnetizations; 16, Murdoch, Thompson Lake and Menihek metavolcanics (1816); 17, Nipissing 'E' magnetizations (magnetization age not established but considered to be 'Hudsonian'); 18, Seward Sub-group; 19, Martin Formation (1930–1635); 20, Flin Flon secondary (magnetization age not established but considered to be 'Hudsonian'); 21, Flin Flon tertiary (1700–1600); 22, Et-Then Group (1835–1630); 23, Sparrow dykes (1700); 24, Menihek 'B' magnetizations; 25, Dubawnt Group (1787); 26, Nonacho sediments (>1700); 27, Seward 'B' magnetizations (magnetization age not established but considered to be 'Hudsonian'); 28, Daly Bay 'A' magnetizations (1622); 29, Cape Smith metavolcanics (1650–1460); 30, 31, Itivdleq area dykes; 32, Kahocella secondary; 33, Metamorphosed Kaminak dykes (1892–1615); 34, Castignon Complex; 35, Argmagssalik geneisses (1800–1600); 36, Kangamuit dykes (1790–1620); 37, Godhåb gneisses; 38, Nordre Stromfjord basement; 39. Holsteinborg basement; 40, Retty 'B' magnetizations; 41, Sagdlerssuaq basement (1620); 42, Ketilidian metavolcanics (1730–1600); 43, W. Greenland amphibolite boudins; 44, Sarfangaqualand amphibolites; 45, Wind River dykes, Wyoming (1880–1680); 46, Eskimo Volcanics, primary (1750–1625); 47, Flaherty Volcanics (1693–1640); 48, Haig intrusions (1790–1620); 49, Eskimo Volcanics, secondary; 50, Sherman Granite; 51, Harp Complex (1450); 52, Michikamau anorthosite (1450–1400); 53, St Francois rocks (1370); 54, Croker Island Complex (1440).

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where they yield a 'B' remanence (Piper 1979 and Fig. 8). These directions are also represented by site 7 of Beckmann (1976) in a dolerite dyke yielding metamorphic K-Ar ages of 1860 and 1440 Ma (Evans & Tarney 1964) and an Rb-Sr age of 1630 Ma (Chapman 1979); the latter age agrees closely with ages derived by Giletti, Moorbath & Lambert (1961) for a pegmatite cutting the dyke and provides an approximate estimate for the age of magnetization. A loosely defined group of directions from the Central Zone denoted 'C' and predominantly of low blocking temperature origin are plotted as antiparallel directions in Fig. 8 and are presumed to be a later part of the apw curve. They may be of composite origin and include Palaeozoic burial PTRM's (see Stewart & Irving 1974); however, the identification of comparable shallow southerly directions from Lewis and Harris (Section 3.5) suggests that a later period of Precambrian apw may be recorded by the Laxfordian terrains of the Lewisian Complex.

5 Middle Proterozoic apw path for the Laurentian Shield

The configuration of the Hebridean craton in the Laurentian Shield is defined by palaeopoles from the Upper Proterozoic Torridonian sediments overlying the Lewisian metamorphic rocks. Both the positions and time-sequence of these poles (Smith *et al.* 1981) match the post-1100 Ma apw path for the Laurentian Shield on the conventional Bullard, Everett & Smith (1965) reconstruction of the North Atlantic continents; the match is improved however, if the Hebridean region is rotated 8° clockwise about a local axis of rotation (Smith *et al.* 1982). If this operation is performed on the mean Lewisian palaeopoles, their positions relative to the Laurentian Shield are as listed below. This operation is equivalent to a clockwise rotation of 38° about a Euler pole at 27.7° E, 88.5° N, and a clockwise rotation of 8° about a Euler pole at 354° E, 57.5° N.

	Palaeomagnetic pole	Pole relative to the Laurentian shield
A_1	42.3 S 239.0 E (antipole)	40.2 S 198.5 E
A_2	37.6 N 273.2 E	41.9 N 225.9 E
A_3	48.9 N 230.2 E	49.8 N 179.4 E
A_4	28.4 N 228.6 E	29.3 N 181.0 E
A_5	16.2 N 224.3 E	16.9 N 178.2 E
B	2.1 S 233.1 E	2.7 S 195.3 E

Apw motion relative to the Laurentian Shield during the interval covered by the tectonism and uplift of the Hudsonian mobile belts has recently been discussed by a number of authors (Cavanaugh & Seyfert 1977; Irving & McGlynn 1979; Piper 1979, 1980) and need not be covered in detail here. The record of palaeopole positions is summarized in Fig. 9. It commences with the *ca.* 1900–1800 Ma record from the Coronation geosyncline and Authapuscow aulacogen (Irving & McGlynn 1979) in the NW sector of the Shield defining a path which incorporates poorly dated poles from elsewhere in the Shield; the high blocking temperature A1 remanence yields an antipole close to this track. The path continued through a sequence of poles representing deformation and uplift events of the Hudsonian terrains *ca.* 1800–1700 Ma in age. Part of this trend is well-defined by the sequence of uplift magnetizations in the Nagssugtoqidian terrains of Greenland (Morgan 1976a; Beckmann 1979) which overlaps in part with the record from the Churchill terrain of the Canadian Shield, but continues through a sequence of poles from amphibolite facies rocks of the Nagssugtoqidian and Ketilidian terrains of Greenland (Beckmann & Mitchell 1976; Piper & Steam 1976; Piper 1980) and all dated somewhat younger than 1700 Ma. The Lewisian poles A_2–A_5 follow

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sequentially on from the youngest part of this track consistent with a likely upper age of 1650–1625 Ma.

Until recently this part of the polar path was unrepresented by data from the Hudsonian terrains of the Canadian Shield. Schmidt (1980) however, finds nearly identical remanence directions from the igneous rocks of Belcher Island, Northwest Territories, where deformation of these rocks at 1700–1600 Ma is correlated with the Hudsonian episode. The remanence of the Eskimo Volcanics (K-Ar age 1625 ± 21 Ma, Rb/Sr age 1750 Ma) is close to the Lewisian A_3 remanence, while poles from the younger Flaherty Volcanics and Haig intrusives follow sequentially from the A_3 to A_5 poles and suggest the possible return path of a large polar loop (Fig. 9). The polar path is not again reasonably well defined until *ca.* 1450 Ma with poles from the anorthosite and related bodies of this age at *ca.* 210° E, 0° N. Efforts to define the intervening path will need to concentrate on the youngest uplift and low blocking temperature components in the Hudsonian terrains; our preliminary results from the Outer Hebrides suggest the possible significance of that area in this context. The age assignments to the Belcher Island rocks and the pole positions of the *ca.* 1450 Ma units suggest that the Lewisian track A2–A5 is appreciably older than 1450 Ma and may represent less than 100 Ma at *ca.* 1600 Ma. This however is not in complete agreement with the geological history because the shallow directions from the Torridon area postdate the LD3 folding estimated to be *ca.* 1599 Ma in age. There are two possibilities: (1) either LD3 is older than indicated by the K-K-Ar data, or (2) these shallow directions belong to a younger part of the apw path than that defined by the other Lewisian and Belcher Island data.

Another remaining problem concerns the A_1 remanence: if this follows sequentially on the A_2–A_5 record as indicated in Fig. 8, then it does not correlate with any known Laurentian data. As noted above, the antipole lies close to the outer part of the loop defined by the Coronation rocks (6 and 7, Fig. 9). It is possible therefore, that the A_1 remanence records a stage during the Hudsonian tectonism at *ca.* 1800 Ma, rather than a later uplift, and that there is no direct link between the A_1 and A_2 remanences as suggested in Fig. 8, however, further studies will be necessary to resolve this point.

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